

Project Name:**Library Management System****Group Members:**

- Zenab Noor BSAI51S25R001
- Saba Maryam BSAI51S25R005
- Shahzeen Fatima BSAI51S25R009

Functional Requirements

Functional requirements includes

1 Book Management

- 1) The system shall allow librarians to add, update, search, and remove book records.
- 2) Each book shall have a unique identification number.
- 3) The system shall store complete book details, including title, author(s), ISBN, publication date, publisher, version, and serial number.
- 4) The system shall maintain and track the number of available copies of each book.

2 User Management

- 1) The system shall maintain separate profiles for students and teachers/admins.
- 2) The system shall allow authorized personnel to create, update, and manage user accounts.
- 3) Student records shall include personal and academic information such as name, DOB, class, session, contact details, and address.
- 4) Teacher and admin accounts shall include credentials and access permissions based on their roles.

3 Book Issuing

- 1) The system shall allow books to be issued to students, teachers, and administrators.
- 2) The system shall ensure that a specific book copy can only be issued to one user at a time.
- 3) The system shall enforce borrowing limits based on user type.
- 4) The system shall automatically record issue dates and due dates.

4 Book Return Processing

- 1) The system shall allow users to return borrowed books.
- 2) The system shall update book availability immediately after a return is processed.
- 3) The system shall maintain a history of issued and returned books.

5 Borrowing Rules Enforcement

- 1) The system shall restrict students to a maximum of 5 active book loans.
- 2) The system shall restrict teachers and administrators to a maximum of 8 active book loans.
- 3) The system shall assign a borrowing period of 10 days for students and 20 days for teachers/admins.

6 Reservation Management

- 1) The system shall allow users to reserve books that are currently unavailable.
- 2) The system shall maintain a reservation queue and allocate books according to reservation priority.
- 3) Reserved books shall not be available for general checkout until the reservation is fulfilled or cancelled.

7 Reminder and Notification System

- 1) The system shall automatically send reminders to users two days before the due date of a borrowed book.
- 2) The system shall notify users about reservation availability and overdue books.

8 Fine Management

- 1) The system shall automatically calculate overdue fines at the rate of 5 rupees per day.
- 2) The system shall maintain a record of outstanding and paid fines for each user.
- 3) The system shall generate fine reports when required.

9 Search and Inquiry Services

- 1) The system shall provide search functionality for books using title, author, ISBN, publisher, or other details.
- 2) Users shall be able to view book availability and reservation status.

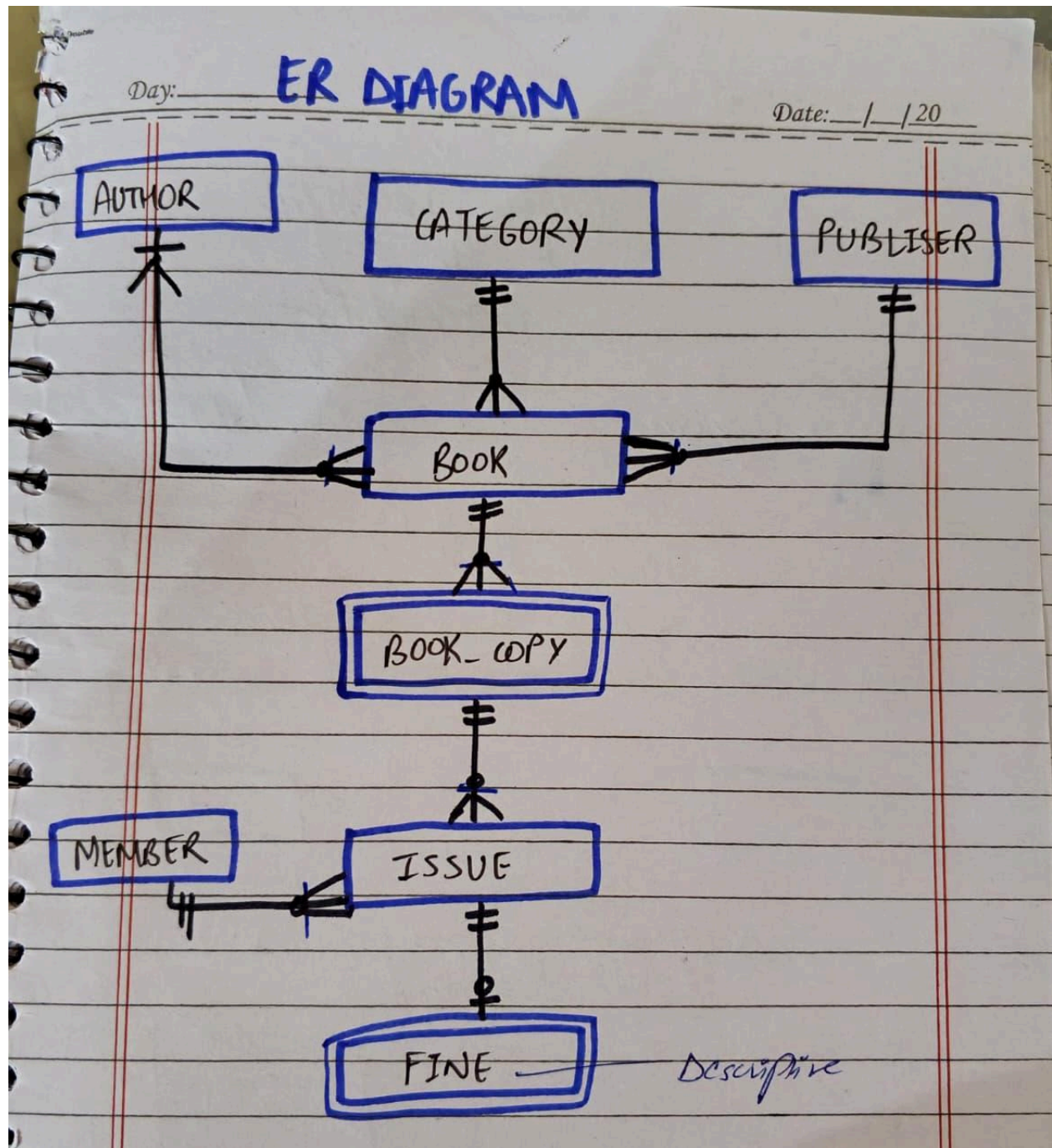
10 Reporting and Record Keeping

- 1) The system shall generate reports related to issued books, returned books, overdue items, reservations, and fines.
- 2) The system shall maintain transaction records for auditing and tracking purposes.

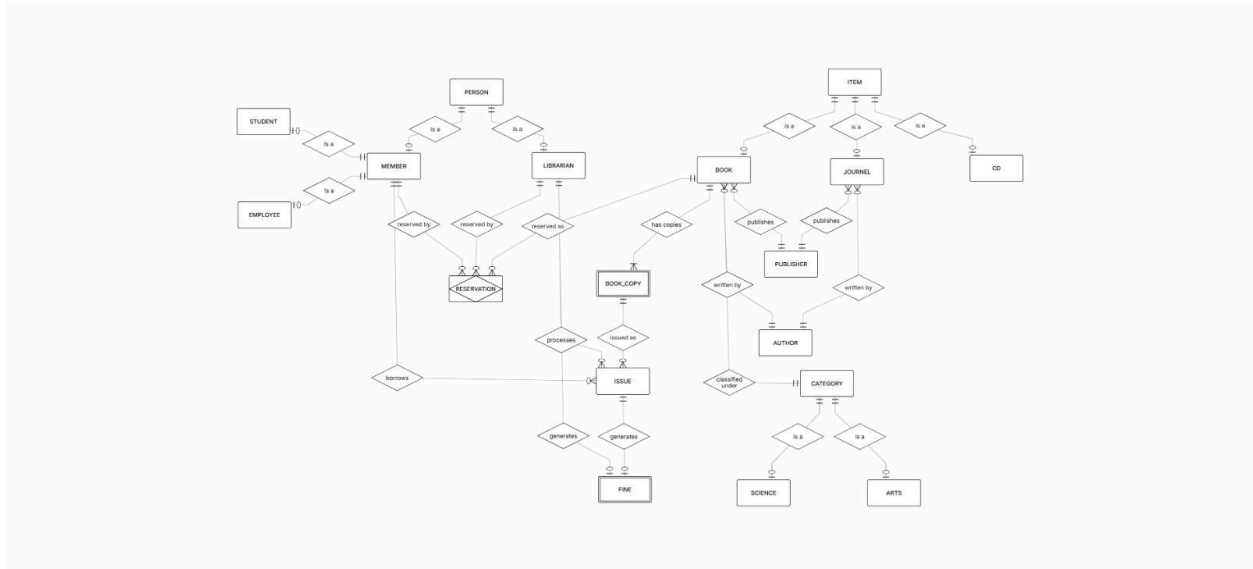
11 Authentication and Access Control

- 1) The system shall require users to log in using authorized credentials.
- 2) The system shall provide different access levels for students, teachers, librarians, and administrators.
- 3) The system shall protect sensitive user and library data from unauthorized access.

ER Diagram



EER DIAGRAM



Entity-Attributes

Entity	Attributes	Description
BOOK	ISBN (PK), TITLE, AUTHOR, STOCK	Represents the inventory of books available.
MEMBER	MEMBER_ID (PK), MEMBER_NAME	Represents the individuals registered to borrow books.
TRANSACTIONS	ISSUE_ID (PK), ISBN (FK), COPY_ID, MEMBER_ID (FK), MEMBER_NAME, ISSUE_DATE, TRANSACTION_TYPE	Logs the movement of books (Issues and Returns).

```
CREATE DATABASE projectdb;  
USE projectdb;
```

```
-- BOOK TABLE  
CREATE TABLE BOOK (  
    ISBN VARCHAR(20) PRIMARY KEY,  
    TITLE VARCHAR(100),  
    AUTHOR VARCHAR(100),  
    STOCK INT  
);
```

```
-- MEMBER TABLE  
CREATE TABLE MEMBER (  
    MEMBER_ID INT PRIMARY KEY,  
    MEMBER_NAME VARCHAR(100)  
);
```

```
-- TRANSACTIONS TABLE  
CREATE TABLE TRANSACTIONS (  
    ISSUE_ID INT AUTO_INCREMENT PRIMARY KEY,  
    ISBN VARCHAR(20),  
    COPY_ID INT,  
    MEMBER_ID INT,  
    MEMBER_NAME VARCHAR(100),  
    ISSUE_DATE DATE,  
    TRANSACTION_TYPE VARCHAR(20),  
    FOREIGN KEY (ISBN) REFERENCES BOOK(ISBN),  
    FOREIGN KEY (MEMBER_ID) REFERENCES MEMBER(MEMBER_ID)  
);
```

```
-- INSERT BOOKS  
INSERT INTO BOOK (ISBN, TITLE, AUTHOR, STOCK) VALUES  
( '978-01', 'Atomic Habits', 'James Clear', 10),  
( '978-02', 'The Alchemist', 'Paulo Coelho', 6),  
( '978-03', 'Rich Dad Poor Dad', 'Robert Kiyosaki', 4),  
( '978-04', 'Think and Grow Rich', 'Napoleon Hill', 8),  
( '978-05', 'The Psychology of Money', 'Morgan Housel', 5);
```

```
-- INSERT MEMBERS  
INSERT INTO MEMBER (MEMBER_ID, MEMBER_NAME) VALUES  
(101, 'Ali'),  
(102, 'Ayesha'),  
(103, 'Ahmed'),
```

```
(104, 'Fatima'),  
(105, 'Hassan');
```

```
-- INSERT TRANSACTIONS
```

```
INSERT INTO TRANSACTIONS
```

```
(ISBN, COPY_ID, MEMBER_ID, MEMBER_NAME, ISSUE_DATE, TRANSACTION_TYPE)  
VALUES
```

```
('978-01', 1, 101, 'Ali', '2025-05-15', 'ISSUE'),  
( '978-02', 1, 102, 'Ayesha', '2025-05-16', 'ISSUE'),  
( '978-01', 1, 101, 'Ali', '2025-05-20', 'RETURN'),  
( '978-03', 1, 103, 'Ahmed', '2025-05-22', 'ISSUE'),  
( '978-04', 1, 104, 'Fatima', '2025-05-25', 'ISSUE');
```